
Schwarz-Christoffel Integrals and Numerical Methods

Computation of Conformal Mappings

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Geometric Foundation

A conformal map is locally equivalent to a **rotation** followed by a **scaling**.

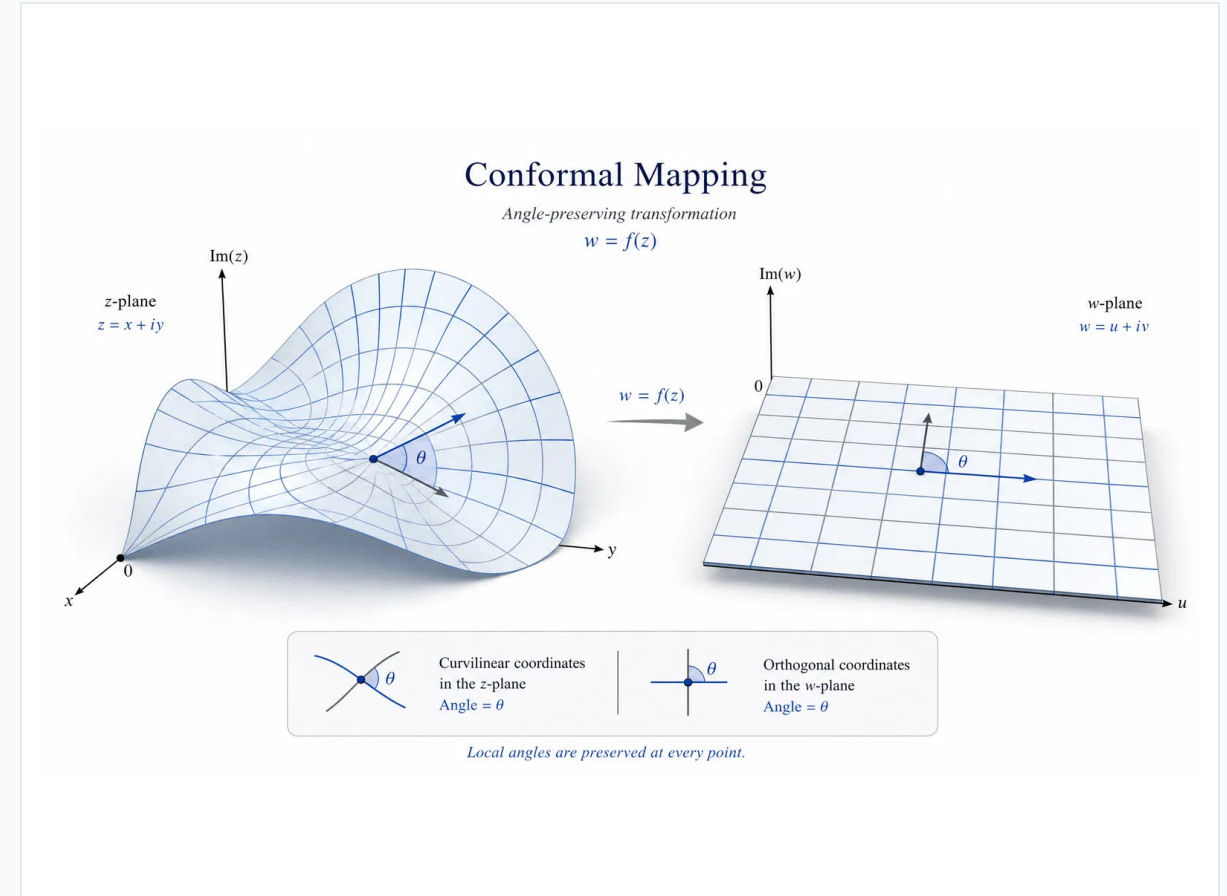
Scaling: $|f'(z)|$ Rotation: $\arg f'(z)$

Conformal Property

If $w = f(z)$ is analytic and $f'(z) \neq 0$, infinitesimal figures are rotated and scaled but not sheared, preserving local angles.

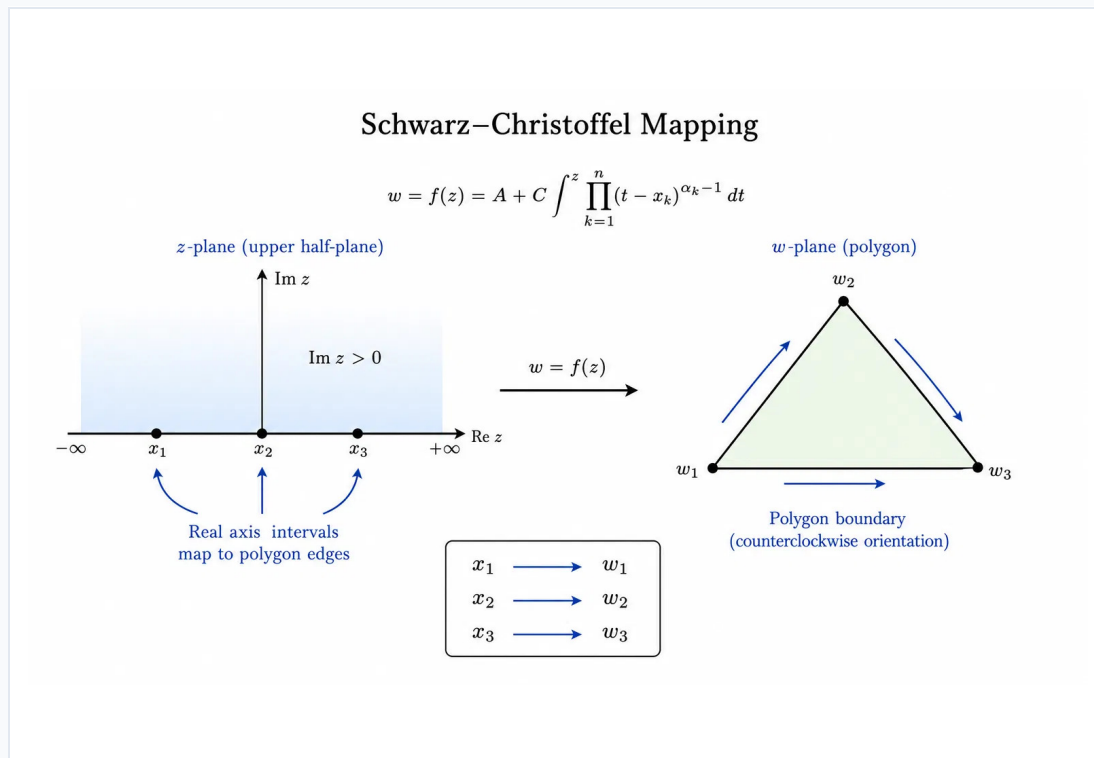
Core Utility

Transforms complex physical boundaries into canonical domains where Laplace's equation $\nabla^2 \phi = 0$ is easier to solve.



Schwarz-Christoffel Mapping Revisited

$$w(z) = C_1 \int^z \prod_{k=1}^n (\zeta - x_k)^{-\alpha_k} d\zeta + C_2$$



Angle Parameters

Defined as $\alpha_k = 1 - \theta_k/\pi$, where θ_k is the interior angle at the vertex w_k . The sum condition $\sum \alpha_k = 2$ ensures a closed polygon.

Boundary Correspondence

The real axis in the preimage plane is mapped to the polygon boundary. Prevertices x_k produce the required turns at the vertices.

Argument Change

As z crosses x_k , the argument of dw/dz changes by $\alpha_k\pi$, which is precisely the exterior turning angle required at the polygon vertex.

The Numerical Parameter Problem

The Core Issue

While polygon angles are fixed by the exponents α_k , the **relative side lengths** are determined solely by the spacing of the prevertices x_k on the real axis.

Möbius Normalization

Three prevertices can be fixed arbitrarily (e.g., $-1, 0, 1$), leaving $n - 3$ essential unknowns to be determined numerically.

Residual Equations

$$F_j(X) = I_j(X) - \lambda_j I_1(X) = 0$$

Where I_j is the j -th side integral and λ_j is the target side-length ratio.

Degrees of Freedom

For an n -vertex polygon, we solve a system of $n - 3$ nonlinear equations to match the target geometry precisely.

Geometric Constraint

The prevertices must maintain their relative ordering $x_1 < x_2 < \dots < x_n$ throughout the numerical solution process.

Solving the Parameter Problem: Newton Iteration

$$\mathbf{x}^{(m+1)} = \mathbf{x}^{(m)} + \eta \mathbf{h}^{(m)}, \quad \mathbf{J}(\mathbf{x}^{(m)}) \mathbf{h}^{(m)} = -\mathbf{F}(\mathbf{x}^{(m)})$$

Linearization & Jacobian

The residual vector $\mathbf{F}$ is linearized at each step. The Jacobian entries $J_{j\nu} = \frac{\partial I_j}{\partial x_\nu} - \lambda_j \frac{\partial I_1}{\partial x_\nu}$ are computed analytically or via finite differences.

Damped Step Constraints

To maintain prevertex ordering ($x_1 < x_2 < \dots < x_n$), a damping factor $0 < \eta \leq 1$ is applied if a full Newton step violates the geometry or increases the residual.

Stopping Criteria

Iteration terminates when both the residual norm $\|\mathbf{F}\|$ and the update norm $\|\mathbf{h}\|$ fall below a specified tolerance ε , typically 10^{-8} .

Numerical Stability

Accuracy depends on the precise evaluation of side integrals I_j . Crowded prevertices can lead to ill-conditioned Jacobians, requiring robust normalization.

Numerical Integration: Singularity Removal

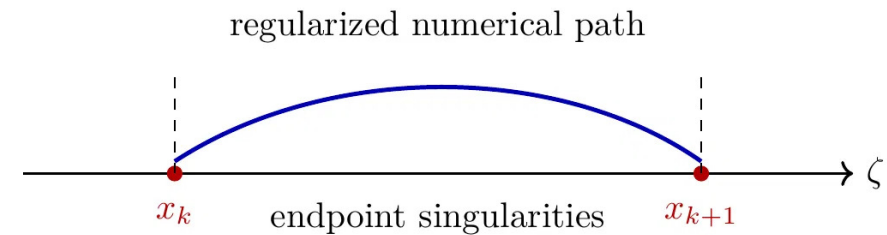
The Singularity Challenge

A side integral has singular behavior at both prevertices x_k and x_{k+1} . Standard quadrature rules fail as the integrand becomes infinite at these endpoints.

$$F(\zeta) = \frac{A_k}{(\zeta - x_k)^{\alpha_k}} + \frac{B_{k+1}}{(x_{k+1} - \zeta)^{\alpha_{k+1}}} + R(\zeta)$$

Kantorovich Regularization

Leading singular terms are integrated analytically, while the smooth remainder $R(\zeta)$ is evaluated using quadrature rules adapted to algebraic endpoint singularities.



Extremal Principles: Minimizing Principle

Variational Characterization

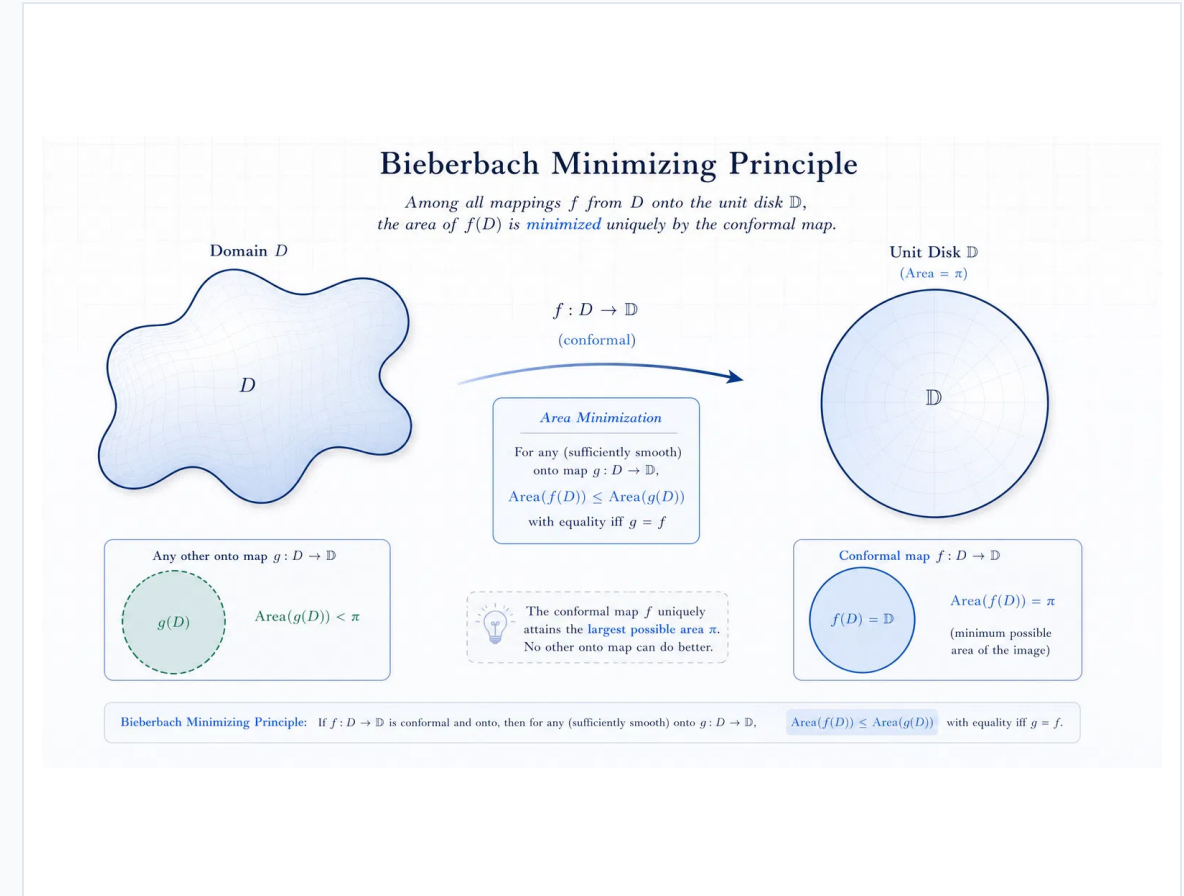
Characterizes the conformal mapping of a simply connected region D onto a disk as the solution of a minimization problem in a Hilbert space of analytic functions.

$$\text{Minimize } \iint_D |f'(z)|^2 dS_z$$

in the class $K_1 = \{f \in L^2(D) : f \text{ is analytic, } f'(a) = 1\}$

Area Minimization

The Riemann mapping theorem guarantees that the derivative of the mapping function is the unique solution, achieving the minimum area πR^2 .



The Bergman Kernel Method

REPRODUCING PROPERTY

$$\iint_D K(z, a) f(z) dS_z = f(a), \quad \forall f \in L^2(D)$$

Orthonormal Basis Construction

The kernel $K(z, a)$ is approximated through a complete orthonormal system $\{\phi_j(z)\}$ in the analytic $L^2(D)$ space:

$$K(z, a) = \sum_{j=1}^{\infty} \overline{\phi_j(a)} \phi_j(z)$$

Conformal Map Recovery

Once the kernel is approximated (e.g., via Gram-Schmidt), the mapping function $f(z)$ is recovered by integration:

$$f(z) = \frac{1}{K(a, a)} \int_a^z K(\zeta, a) d\zeta$$

Applications: Potential Theory & Fluid Flow

Potential Theory

Conformal maps transfer potential fields (electrostatics, heat) between complex and simple domains, preserving the harmonic nature of solutions to $\nabla^2\phi = 0$.

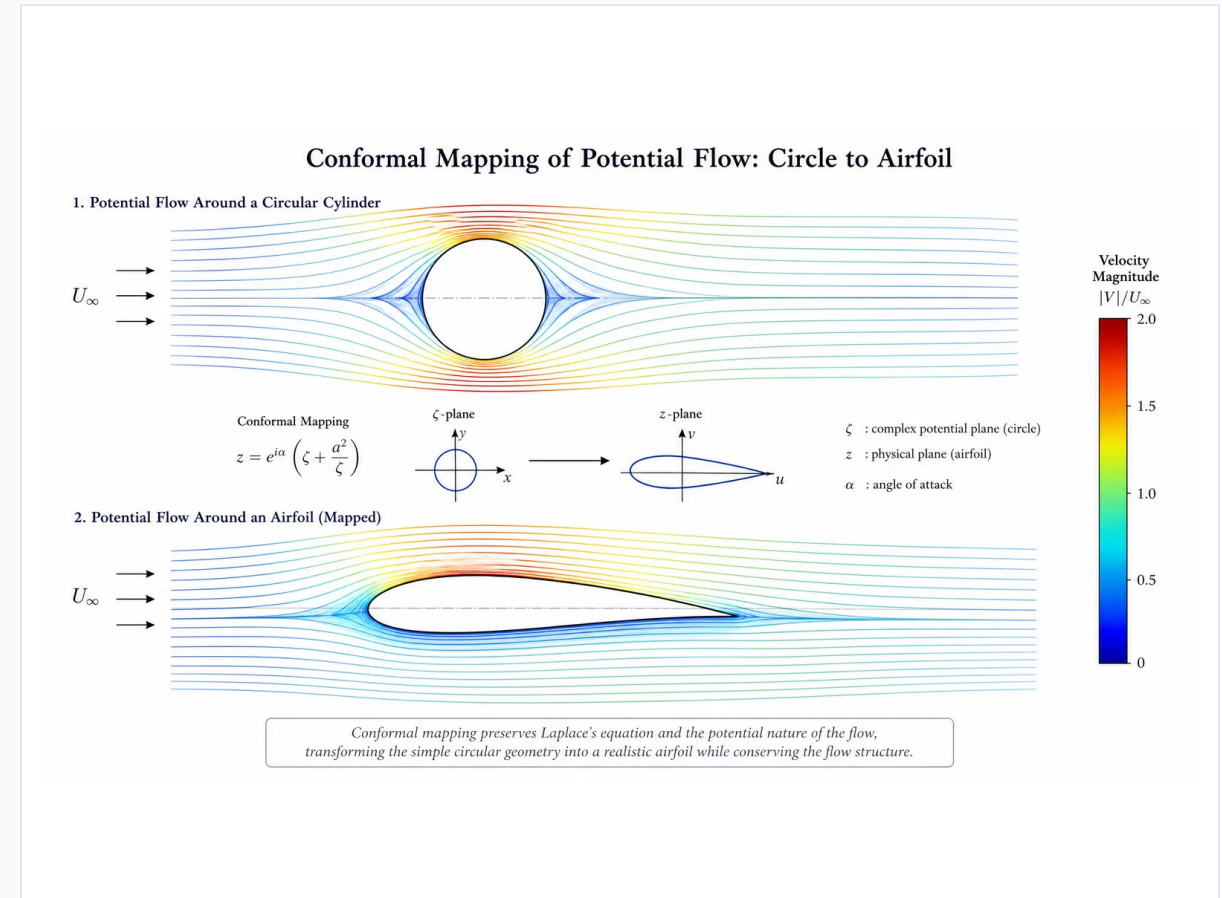
$$F(z) = \phi(x, y) + i\psi(x, y)$$

Fluid Flow Analysis

The complex potential F combines velocity potential ϕ and stream function ψ . Streamlines and equipotentials remain orthogonal under mapping.

Aerodynamic Mapping

Flow patterns around complex obstacles, such as airfoils, are derived by mapping canonical flows around circular cylinders.



Grid Generation & Conclusion

COMPUTATIONAL GRID GENERATION

- Conformal mapping naturally creates **boundary-fitted orthogonal coordinate curves**, essential for high-accuracy numerical simulations.
- Preserving orthogonality simplifies transformed differential operators, improving the stability of CFD and structural analysis solvers.
- Useful for mesh generation in complex polygonal regions where standard rectangular grids fail to align with physical boundaries.

PROJECT SUMMARY

- The SC formula provides the angular structure, while Newton iteration and singularity subtraction solve the numerical construction.
- Extremal methods via the Bergman kernel offer a powerful variational alternative for general simply connected domains.
- Practical relevance spans potential theory, fluid flow, and heat transfer by simplifying complex domain geometries.

Thanks!!